

Table 3: Learning objectives (Theory) of Course

Paper 1									
A3 Course outcome	B3 Learning Objective (At the end of the session, the students should be able to)	C3 Doma in/sub	D3 Must to know / desirable to know / Nice to know	E3 Level Does/ Shows how/ Knows how/ Know	F3 T-L meth od	G3 Assessment (Refer abbreviations)	H3 Form ative/ summ ative	I3 Term	J3 Integr ation
Topic 1 Concepts of Agada Tantra (Clinical Toxicology) (Lecture :8 hours, Non lecture: 1 hours)									
CO1	Define Agada Tantra and Toxicology. Describe the scope of Agada Tantra in the present era.	CK	MK	K	L&PP T	T-EMI,T- EW	F&S	I	
CO1	Define visha and poison. Enlist its synonyms.	CK	MK	K	L&PP T	T-EMI	F&S	I	
CO1	Differentiate between poison,venom and toxin.	CK	MK	K	L,L& PPT	P-VIVA	F&S	I	
CO1	Describe classification of visha & poison	CK	MK	K	L&PP T	T-EMI,P- VIVA	F&S	I	
CO1	Describe sthavara and jangama visha adhisthana.	CK	MK	K	L&PP T	T- EW	F&S	I	
CO1	Describe the routes of administration of poison.	CC	MK	K	L&PP T	T- EW,P- VIVA	F&S	I	
CO1	Enlist visha guna and differentiate between visha, madya and oja	CK	MK	K	L&PP	T-EMI	F&S	I	

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CO1	Describe the mode of action of visha (visha gunanusar karma) & poison (Introduction to toxicokinetics).	CC	MK		KH	L,L&PPT,L&GD	T-EMI,T- EW	F&S	I	
CO1	Describe the factors modifying the action of poison.	CC	MK		KH	L&PP T	T- EW	F&S	I	
CO1	Describe vishavardhaka bhava and visha sankata.	CC	MK		KH	L&PP T	T-EMI,T- EW,P-VIVA	F&S	I	
CO1	Describe visha vega & vegantara	CK	DK		K	L&PP T	T-EMI,P- VIVA	F&S	I	
CO1	Describe sthavara visha veganusara lakshana and chikitsa	CK	MK		K	L,L&PPT	T-EMI	F&S	I	
CO1	Describe vishapeeta and vishamukta lakshana.	CK	MK		K	L&G D	T-EMI,P- VIVA	F&S	I	
Topic 2 Visha Chikitsa (Management of Poisoning) (Lecture :5 hours, Non lecture: 4 hours)										
CO2	Describe the diagnosis of poisoning in living and dead.	CAP	MK		KH	L&PP T	T-EMI,T- EW,P-VIVA	F&S	I	
CO2	Enumerate and discuss chaturvimshati upakrama	CC	MK		KH	L&PP T	T- EW,P- VIVA	F&S	I	
CO2	Describe the general principles of managment of poisoning.	CAP	MK		KH	L&PP T	T-EMI,P- VIVA	F&S	I	V-KC
CO2	Describe the duties of medical officer in case of suspected poisoning.	CK	MK		K	L&PP T	T-EMI,P- VIVA	F&S	I	
Topic 3 Vishakta aahara pariksha and Viruddha ahara (Lecture :3 hours, Non lecture: 2 hours)										

CO1,CO2,CO 3	Explain the various sources of exposure of visha like anna, paana, vastra, abhyanga, lepa, paduka, abharana, etc.	CK	DK	K	L&PP T,L& GD	T- EW,CL- PR,M-CHT	F&S	I	
CO1,CO2,CO 3	Describe the Ayurvedic methods of detection of food and beverages contaminated with visha.	CC	MK	KH	L&PP T,L_ VC,S DL,D	T- EW,P-VIV A,P-EXAM	F&S	I	
CO1,CO2,CO 3	Define adulteration, describe the methods of detection of adulterants and its relevance in Agada Tantra.	CAP	DK	KH	L_ VC ,SDL, PT	T- EW,P-VIV A,P- EXAM,OSPE ,RK	F&S	I	
CO1,CO2,CO 3	Describe the modern analytical techniques for the detection of poison (in contaminated articles) like chromatography, spectroscopy, etc.	CC	NK	KH	L&PP T,L& GD,L _VC	T- EW,P- VIVA,CL-PR	F&S	I	
CO1,CO2,CO 3	Define viruddha ahara and explain its types.	CC	MK	KH	L&PP T,L& GD,B S	T- EW,P- VIVA,INT	F&S	I	
CO1,CO2,CO 3	Discuss and illustrate the contemporary importance of viruddha ahara with examples	CAN	MK	KH	L&G D,BS, SDL	T- EW,QZ ,CL-PR	F&S	I	
CO1,CO2,CO 3	Discuss the application of concepts of viruddha ahara in the prevention and management of diseases.	CAP	MK	KH	L&G D,BS, SDL	T- EW,P-VIV A,CR-RED	F&S	I	
CO1,CO2,CO 3	Describe food poisoning, its types and management.	CC	MK	KH	L&PP T,L&	T- EW,P-VIV A,CR-RED	F&S	I	

CO3,CO4	Describe the clinical presentations and diagnosis of paduka visha, abharana visha, vastra visha, lepa visha, abhyanga visha, etc.	CAP	MK	KH	L&PP T,DIS ,CBL, CD	T- EW,P- CASE,CL-PR	F&S	II	
CO3,CO4	Discuss the management of paduka visha, abharana visha, vastra visha, lepa visha, abhyanga visha, etc.	CAN	MK	KH	L&PP T,DIS ,CBL, PER	T- EW,P-EXA M,P- CASE,CL-PR	F&S	II	
CO3,CO4	Describe contact dermatitis and its types.	CK	MK	K	L&PP T	T- EW,P- VIVA	F&S	II	
CO3,CO4	Discuss the application of treatment principles of Agada Tantra in the contact dermatitis.	CAP	MK	KH	L&PP T,CB L	P-VIVA,P- CASE,CL-PR	F&S	II	
CO3,CO4	Discuss the dermatological conditions caused due to garavisha and dooshivisha.	CAP	MK	KH	L&PP T,DIS ,CBL, CD	T- EW,P-VIV A,P-CASE	F&S	II	
CO3,CO4	Discuss the application of principles of treatment of garavisha and dooshivisha in the management of dermatological manifestations.	CAP	MK	KH	L&PP T,DIS ,CBL	P-VIVA,P- CASE,CL-PR	F&S	II	
CO3,CO4	Discuss the daignosis and application of the treatment principles of Keeta visha in dermatological manifestations due to bites & stings.	CAP	MK	KH	L&PP T,DIS ,CBL, SDL	T- EW,P-VIV A,P-CASE,CL- PR	F&S	II	
Topic 8 Therapeutic utility of Agada yoga (Lecture :1 hours, Non lecture: 0 hours)									
CO1,CO2,CO	Enlist ingredients and discuss the therapeutic utility of	CAP	MK	KH	L&PP	T- EW,P-	F&S	II	

3,CO4	Doohivishari Agada, Bilwadi Agada, Dashanga Agada, Murvadi agada and Panchashririsha Agada.											
CO1,CO2,CO3,CO4	Enlist and discuss the ingredients of vishaghna mahakashaya (Charaka Samhita) and ekasara gana (Sushruta Samhita).	CAP	MK		KH	L&PP T,BS, CBL, FC	T- EW,P- VIVA	F&S	II			
Topic 9 Sthavara visha – Poisons of Plant origin (Lecture :4 hours, Non lecture: 2 hours)												
CO2,CO4,CO6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of vatsanabha.	CAP	MK		KH	L&PP T	T- EW,P- VIVA,P-ID	F&S	II			
CO2,CO4	Mention therapeutic dose and enlist two important formulations of vatsanabha.	CK	NK		K	L	T-EMI,P-VIV A,P-EXAM	F&S	II			
CO2,CO4,CO6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of karaveera.	CAP	MK		K	L&PP T	T- EW,P- VIVA,P-ID	F&S	II			
CO2,CO4	Mention therapeutic dose and enlist two important formulations of karaveera.	CK	NK		K	L	T- EW,T- ME Qs,P-VIVA,P- EXAM	F&S	II			
CO2,CO6	Describe mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of digitalis.	CAP	NK		K	L&PP T	T- EW,P- VIVA	F&S	II			
CO2,CO6	Describe mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of tobacco.	CK	MK		K	L&PP T	T- EW,P-ID	F&S	II			
CO2,CO6	Describe mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and	CK	DK		K	L&PP T	T- EW	F&S	II			

	medicolegal aspects of Cerbera odollam.												
CO2,CO4,CO6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of Kupeelu.	CK	MK	KH	L&PP T	T- EW,P- VIVA,P-ID	F&S	II					
CO2,CO6	Mention therapeutic dose and enlist two important formulations of kupeelu.	CK	NK	K	L	T-EMI	F&S	II					
CO2,CO4,CO6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of ahipheha.	CK	MK	KH	L&PP T	T- EW,P- VIVA,P-ID	F&S	II					
CO2,CO6	Mention therapeutic dose and enlist two important formulations of ahipheha.	CK	NK	K	L	T-EMI	F&S	II					
CO2,CO4,CO6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of dhatura.	CK	MK	KH	L&PP T	T- EW,P- VIVA,P-ID	F&S	II					
CO2,CO6	Mention therapeutic dose and enlist two important formulations of dhatura.	CK	NK	K	L	T-EMI	F&S	II					
CO2,CO4,CO6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of Bhanga.	CK	MK	K	L&PP T	T- EW,P-VIV A,P-EXAM,P- ID	F&S	II					
CO2,CO6	Mention therapeutic dose and enlist two important formulations of bhanga.	CK	NK	K	L	T-EMI	F&S	II					
CO2,CO4,CO6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of jayapala.	CK	MK	K	L&PP T	T- EW,P- VIVA,P-ID	F&S	II					
CO4	Mention therapeutic dose and enlist two important formulations	CK	NK	K	L	T-EMI,P-	F&S	II					

	of jayapala.						EXAM			
CO2,CO4,CO 6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of gunja.	CAP	DK	KH	L&PP T	T- EW,P- VIVA,P-ID	F&S	II		
CO4	Mention therapeutic dose and enlist two important formulations of gunja.	CK	NK	KH	L	T- EW,P-VIV A,P-EXAM	F&S	II		
CO2,CO4,CO 6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of bhallataka.	CAP	MK	KH	L&PP T	T- EW,P- VIVA,P-ID	F&S	II		
CO4	Mention therapeutic dose and enlist two important formulations of bhallataka.	CK	NK	K	L	T-EMI,P-VIV A,P-EXAM	F&S	II		
CO2	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of arka and snuhi.	CK	DK	K	L&PP T	T- EW,P- VIVA	F&S	II		
CO2,CO4,CO 6	Describe the active principles, mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance and medicolegal aspects of langali.	CK	NK	K	L&PP T	T- EW,P- VIVA,P-ID	F&S	II		
Topic 10 Sthavara Visha – Poisons of Metallic origin (Lecture :4 hours, Non lecture: 2 hours)										
CO2	Describe the mode of action, fatal dose, fatal period, signs and symptoms, management, post-mortem appearance, medicolegal aspects of acute poisoning of 1. Arsenic 2. Mercury 3. Lead 4. Copper	CK	MK	K	L&PP T,PB L	T-EMI,T- MEQs,P- VIVA	F&S	II		

	Write its therapeutic dose and enlist any four important formulations.												
CO1,CO2	Discuss the Ayurvedic concepts for the diagnosis and management of chronic heavy metal toxicity.	CC	DK		K	L&PP T,L& GD	T-EMI,T- MEQs,P- VIVA	F&S	II				
CO2,CO3,CO 4	Discuss the toxicological implications of improperly prepared medicines with special reference to those containing metals, minerals and poisonous substances.	CAP	NK		KH	L&PP T,CB L,FC	P-VIVA	F&S	II				
Topic 11 Jangama Visha (Lecture :10 hours, Non lecture: 6 hours)													
CO2,CO5	Describe sarpa bheda and enumerate the identification features of darvikara, mandali and rajimantha sarpa.	CAP	MK		KH	L&PP T,BS, CBL	T- EW,P-VIV A,P-EXAM	F&S	II				
CO2,CO5	Classify snakes and describe the features of venomous and non venomous snakes and identify their bites.	CAN	MK		K	L&PP T	T- EW,P-VIV A,P-EXAM	F&S	II				
CO2,CO5	Describe the fatal dose, fatal period, signs and symptoms of common cobra, king cobra, russel's viper, saw scaled viper, pit vipers and common krait.	CAN	MK		KH	L&PP T,DIS ,BS,C BL	T- EW,T-CS,P -VIVA,P- EXAM	F&S	II				
	Describe the samanya lakshana and samanya chikitsa of darvikara, mandali and rajimanth sarpa damsha.	CK	MK		K	L	T-EMI	F&S	II				
CO2,CO5	Describe the management of snake bites according to recent WHO snake bite management guidelines.	CAN	MK		KH	L&PP T,CB L,PE R,CD	T- EW,P-VIV A,P-EXAM	F&S	II				
CO2,CO5	Describe the classification, diagnosis, samanya lakshana and chikitsa of lootavisha with its contemporary aspects.	CAN	MK		KH	L&PP T,CB	T- EW,P- VIVA	F&S	II				

	management of chronic alcoholism.						D,BS	VIVA				
CO2,CO4	Discuss the signs & symptoms of alcohol withdrawal with its Ayurvedic approach.	CC	MK		KH	L&G D,BS, IBL	T- EW,P- VIVA	F&S	III			
CO2,CO4	Discuss the signs & symptoms of withdrawal of bhang (cannabis), aliphena/opium and its derivatives, tobacco, LSD and MDMA with its Ayurvedic approach.	CC	MK		KH	L&G D,BS, TUT	T- EW,P- VIVA	F&S	III			
CO2,CO4	Discuss the reseach updates in Ayurveda in the management of conditions due to substances of abuse.	CC	MK		KH	L&G D	T-EMI,T- ME Qs,T-CRQs,P- VIVA	F&S	III			
CO7	Describe the NDPS Act.	CK	DK		K	L&PP T	T- EW	F&S	III			
Topic 14 Agada Tantra perspectives on cancer (Lecture :2 hours, Non lecture: 4 hours)												
CO2,CO4	Discuss the concepts of Agada Tantra in oncology.	CC	MK		KH	L&G D,BS, PBL	T-EMI,T- MEQs,P- VIVA	F&S	III			
CO1,CO2	Enlist the cancer causing toxic chemicals / substances (toxic carcinogens) and describe the process of mutation in genes and carcinogenesis.	CC	MK		K	L&G D	T-EMI,T- ME Qs,T-CRQs,P- VIVA,SA	F&S	III			
CO2,CO4	Describe the chemotherapy induced common toxicities and Ayurvedic approach for its management.	CC	MK		KH	L&PP T,PB L	T-EMI,T- ME Qs,T-CRQs,P- VIVA,SA	F&S	III			
CO2,CO4	Describe the radiotherapy induced common toxicity and Ayurvedic approach for its management.	CC	MK		KH	L&G D,PB L,SD L	T-EMI,T- ME Qs,T-CRQs,P- VIVA	F&S	III			

CO2,CO4	Discuss the role of visha and vishaghna dravya in the management of cancer along with its research updates.	CC	DK	KH	L&G D,BS	T-EMI,T- ME Qs,T-CRQs,P- VIVA	F&S	III	
Topic 15 Forensic medicine (Vyavahara Ayurveda) and Medical jurisprudence (Vidhi vaidyaka): (Lecture :1 hours, Non lecture: 0 hours)									
CO1,CO6,CO 7	Define Forensic Medicine & Medical Jurisprudence. Give an introduction to Indian Penal Code (IPC), Criminal Procedure Code (CrPC) and Indian Evidence Act (IEA).	CC	MK	KH	L&G D	T- EW,P- VIVA,QZ	F&S	I	
Topic 16 Vaidya sadvritta : Duties and Responsibilities of medical practitioner (Lecture :6 hours, Non lecture: 4 hours)									
CO1,CO6,CO 7	Describe vaidya sadvritta, medical ethics, code of conduct, Charaka oath and Hippocratic oath.	CK	MK	K	L&PP T,L& GD,B S,PER	T- EW,P- VIVA	F&S	I	
CO1,CO6,CO 7	Describe the constitution, objectives and functions of NCISM.	CK	MK	K	L&PP T,DIS	T- EW,P- VIVA	F&S	I	
CO6,CO7	Describe the duties and rights and privileges of a registered medical practitioner.	CK	MK	K	L&PP T,DIS	T- EW,P- VIVA	F&S	I	
CO1,CO6,CO 7	Describe consent, professional secrecy and privileged communication.	CC	MK	K	L&PP T,BS	T- EW	F&S	I	
CO1,CO6,CO 7	Describe professional negligence, professional misconduct and unethical practices.	CC	MK	K	L&G D,CB L,PE R	T- EW,P- VIVA	F&S	I	
CO6,CO7	Explain the defenses in medical negligence suits with appropriate case laws.	CAP	MK	K	L&PP T,L& GD,B	T- EW,P- VIVA	F&S	I	

CO6	Define death and describe the stages and modes of death. Explain the immediate changes following death.	CC	MK	KH	L&PP T,L& GD,L _VC	T- EW,P- VIVA	F&S	II	
CO6	Enlist and explain the early changes following death. Describe Algor mortis, rigor mortis and livor mortis with regards to their definition, mechanism, modifying factors and medico legal importance.	CC	MK	KH	L&PP T,L_ VC,D IS,SI M	T- EW,P- VIVA	F&S	II	
CO6	Enlist the late changes following death and describe the definition, modifying factors and medico legal importance of putrefaction, mummification and adipocere formation.	CC	MK	KH	L&PP T,L_ VC,D IS	T- EW,P- VIVA	F&S	II	
CO6	Define Autopsy and describe its types, its objectives, rules and procedure along with exhumation.	CC	MK	KH	L&PP T,L_ VC	T- EW,P- VIVA	F&S	II	
CO6	Describe the Transplantation of Human Organs and Tissues Act.	CC	NK	KH	BS,S DL,P ER	T- EW,P- VIVA	F	II	
Topic 20 Asphyxial deaths (Lecture :4 hours, Non lecture: 2 hours)									
CO6	Define hanging. Describe its classification, causes of death, post-mortem appearances and medico-legal aspects.	CK	MK	KH	L&PP T,DIS	T- EW,T- ME Qs,T-CS,P-	F&S	II	

	narcoanalysis, DNA profiling.																		
Topic 26 Laws, Acts, Rules and Regulations (Lecture :3 hours, Non lecture: 2 hours)																			
CO6,CO7	Describe the 1.NCISM Rules and Regulations in force. 2. Clinical Establishments Act. 3. Medicare Service Persons and Medicare Service Institutions (Prevention of violence and damage or loss to property) Act, 2008. 4. The Drug and Cosmetic Act - schedules related to poison. 5. Rules and Regulations related to AYUSH State Registration Boards of concern state.	CK	DK	K	L&PP T,DIS ,SDL	T- EW, P- VIVA,CL-PR	F&S	III											